

Cellular Intelligence Machine Learning Exercise

This exercise is designed to simulate the types of machine learning challenges we face at Cellular Intelligence . It will help us evaluate your coding ability and serve as a starting point for a deeper discussion during a subsequent interview. Please share your code in any format you prefer.

1. Model Comparison (Discussion Only)

[scDiffusion](#) and [scGPT](#) are two foundational models designed to learn general-purpose representations of cells from RNA data in a self-supervised manner.

- Compare the underlying computational methods used in both models.
- Discuss the advantages and limitations of each approach.

2. Perturbation Modeling

At Cellular Intelligence , we are training machine learning models to predict the effect of signaling perturbation on pluripotent stem cells in order to control cellular differentiation and develop impactful cell therapies. In this exercise, your goal is to train a machine learning model to predict the cumulative effect of a sequence of perturbations on cells.

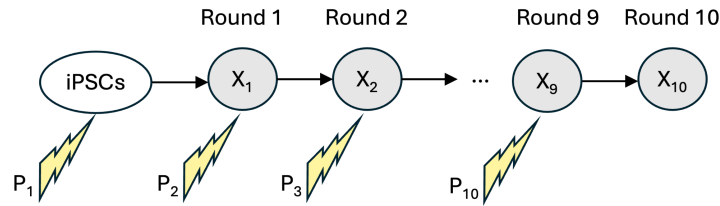
In this experiment, induced pluripotent stem cells (iPSCs) have been exposed to a series of 10 perturbations. After each exposure, we measure each cell's transcriptome, which is a proxy for the cell's state. Thus, this data is comprised of ten tuples:

$$(X_1, P_1), (X_2, P_2), \dots, (X_{10}, P_{10})$$

X_i is a $N \times G$ gene expression matrix (N is the number of cells and G is the number of genes). The same cells have been profiled across all 10 rounds, so row j of X_i is the same cell as row j in X_{i+1} .

P_i is a $N \times C$ perturbation matrix (N is the number of cells and C is the number of compounds included in each perturbation) at round i of the experiment. More specifically, P_i are the perturbations that produced X_i when applied to X_{i-1} . Each row of P_i is a vector of concentrations of the C compounds.

A small schematic diagram of this experiment is shown below (note, we do not have expression data from the original iPSCs, only the resultant gene expression matrix after the first perturbation):



Your goal is to train a model, f , that, given a sequence of perturbations, $P_1, P_2, \dots, P_{10}$, predicts the resultant gene expression matrix, X_{10} , that would result from those perturbations applied sequentially to each cell. More specifically, your task is to train a model:

$$f(P_1, P_2, \dots, P_{10}) \rightarrow X_{10}$$

You will be working with .h5ad files and NumPy files:

- The files, `expression_round_<i>.h5ad`, represent the expression data at each round. This file is an H5ad file.
- The files, `perturbation_round_<i>.numpy`, represent the perturbations at each round. You can load these files via [numpy.load\(\)](#).

Please share an analysis in which you:

1. Implement one (or more) machine learning models for predicting the effect of sequences of perturbations. While the data is a relatively toy dataset (4,807 cells), design your model in such a way that it could theoretically scale to millions of cells.
2. Evaluate model performance
3. Provide a short report that describes the method you implemented, how you evaluated it, and how it performed. How might this model scale with more data? What other modeling approaches would you try out if the data was composed of >100M cells?

Note:

- This data is simulated, toy data that has been embedded down to only 50 latent components. There is no need to assume that these values represent UMI counts. How would you change your framework if these were UMI counts over real genes?

Drive link to files: [link](#)